

Guidance on Initiating and Prescribing Anti-dementia medication in Primary Care

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APPLICABLE LEGISLATION

GLOSSARY OF TERMS

TERM	ACRONYM	DEFINITION
Clinical Commissioning Group	CCG	
Area Clinical Effectiveness Committee	ACE	

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Guidance on Initiating and Prescribing Anti-dementia medication in Primary Care

Background:

This guidance is to be used by Dudley GPs following assessment of patients by the Midland Partnership Foundation Trust, Dementia Assessment Service. After the assessment a letter is sent to the patient's GP outlining the plan of care and any recommendation of Acetylcholinesterase (AChEI) inhibitors.

The guidance can also be used for the discharge of existing patients who have commenced medication treatment from Dudley Walsall Mental Health Trust.

This guidance provides evidence-based approach to prescribing in dementia for GPs and addresses the key questions of when and how to use it.

Key Points:

- **Acetylcholinesterase (AChEI)** inhibitors are recommended for treatment of mild to moderate Alzheimer's disease (AD) mixed vascular dementias (VaD) where there is a significant AD component, mild, moderate and severe Parkinson's dementia (PD) and Dementia with Lewy Bodies (DLB).
- AChEIs currently used in the treatment of dementia include **donepezil, rivastigmine** and **galantamine**.
- **Donepezil is the first-line AChEI of choice in primary care.**
- **Memantine** (NMDA receptor antagonist) is recommended for moderate to severe AD, Mixed vascular dementias with predominant AD component, mild, moderate and severe PD and DLB if intolerant or have contraindications to AChEIs.
- AChEIs or memantine do not reverse the disease progression.
- **None of the above medication is indicated for VaD, Mixed AD with predominant Vascular component or frontotemporal dementia.** Only consider AChEI or memantine for cormorbid VaD with predominant AD, PD or DLB.
- Treatment should **only** be started if a caregiver is available to regularly monitor medication compliance, benefits and side effect.
- Side effects are possible for up to 20% of patients and are greatest on starting therapy or at dose increase, but usually diminish in a few days.
- Patients and carers should be advised not to expect a dramatic reversal of cognitive decline. Modest cognitive and functional benefits reported over 12 -24 months. Not all patients obtain clinical benefit.

1 Check

- The diagnosis of specific dementia is made according to national or locally agreed guidelines.
- **Involve the patient in decisions** and check they understand and have not been coerced into agreeing to treatment.
- **Consider carer/relatives views** (if patient has no mental capacity to consent).
- **Consider co-morbidities** (Table 1) patients age, frailty, benefits versus risks of starting medication.
- **Review current medication and consider potential drug** interactions.
- **AChEI - Check baseline pulse** – See **flow chart** attached **page 5**.
If **bradycardia** is due to rate limiting drugs (beta blockers, calcium channel blockers, digoxin etc.) these could be reviewed and changed before starting AChEIs. If pacemaker is fitted for bradycardia pulse check is not required, prescribe AChEI as usual.
- **Memantine – Check eGFR** – see British National Formulary (BNF) - pulse check not required.

2 Prescribe

- **Prescribe: donepezil tablets 5mg nocte for 4 weeks.** Use orodispersible tablets if the patient has swallowing difficulties or severe behavioural & psychological symptoms.
- **Review patient after 4 weeks.** Increase dose to **10mg nocte if necessary**. Dose can be increased at any stage during treatment if not at 4 weeks. **Maximum daily dose is 10mg**

Renal impairment: no dose adjustment required. Titrate according to individual tolerability.

Hepatic impairment: Mild to Moderate: perform dose escalation according to individual tolerability. No information available for severe impairment.

- **Prescribe: 2nd line Rivastigmine tablets:** 1.5mg twice daily, with the morning and evening meals; increased in steps of 1.5mg twice daily at intervals of at least 2 weeks up to a maximum of 6mg twice daily. Re-titration from 1.5mg twice daily should be undertaken if treatment is interrupted for more than several days. **Liquid and transdermal patch available for patients with swallowing difficulties and severe gastrointestinal side effects (see BNF-British National Formulary).**

Renal impairment: Titrate according to individual tolerability.

Hepatic impairment: Titrate according to individual tolerability in mid to moderate impairment. Use with caution in severe impairment.

2 Prescribe

- **Prescribe: Galantamine** as the **third line** treatment option. **Galantamine: Standard release tablets:** 4mg twice daily for 4 weeks, with the morning and evening meals; increased to 8mg twice daily for 4 weeks. An increase to 12mg twice a day may be considered following assessment including evaluation of clinical benefit and tolerability. If there is no increased response or tolerability is poor, reduce dose back to 8mg twice daily. Ensure adequate fluid intake.

Modified release capsules: Initially 8mg once daily for 4 weeks each morning with food, increased to 16mg once daily for 4 weeks; maintenance 16-24mg once daily (reviewed as per standard tablets).

(MHRA issued a serious skin reaction alert in 2015 in 1:1000 cases. These include Steven Johnson syndrome and acute generalised exanthematous pustulosis. Manufacturer advises discontinue at the first appearance of the skin rash and seek help. Use with caution in unstable angina and congestive cardiac failure).

Renal impairment: Avoid if eGFR is less than 9ml/minute/1.73m²

Hepatic impairment: For immediate release preparations in moderate impairment initially 4mg once daily for at least 7 days then 4mg twice daily for last 4 weeks; max. 8mg twice daily; avoid in severe impairment. For modified release preparations in moderate impairment, initially on 8mg on alternate days for 7 days then 8mg once daily for 4 weeks; max. 6mg daily; avoid in severe impairment.

(If one AChEI is not tolerated, alternative AChEI could be prescribed if it is appropriate taking into account adverse event profile, adherence, comorbidity, drug interactions and dosing profiles).

- **Prescribe: Memantine tablets.**

Memantine monotherapy is recommended as an option for managing moderate to severe AD, PD or DLB who are **intolerant of or have a contraindication to AChEI**.

Memantine tablets 10mg nocte for 4 weeks. Titrate maximum daily dose up to 20 mg per day. In order to reduce the risk of undesirable effects the maintenance dose is achieved by upward titration of 5 mg per week over the first 3 weeks as follows (Starter pack):

Week 1 (day 1-7):

The patient should take half a 10 mg film-coated tablet (5 mg) per day for 7 days.

Week 2 (day 8-14):

The patient should take one 10 mg film-coated tablet (10 mg) per day for 7 days.

Week 3 (day 15-21):

The patient should take one and a half 10 mg film-coated tablet (15 mg) per day for 7 days.

From Week 4 on:

The patient should take two 10 mg film-coated tablets (20 mg) per day.

Maintenance dose

The recommended maintenance dose is 20 mg per day.

Renal impairment: Reduce the dose to 10mg daily if eGFR 30-49ml/minute/1.73m², if well tolerated after at least 7 days dose can be increased in steps to 20mg daily; reduce dose to 10mg daily if eGFR 5-29ml/minute/1.73m². Avoid if eGFR less than 5ml/minute/1.73m²

Hepatic impairment: Avoid in severe impairment. No information available.

(Patients with moderate to severe AD who are already on AChEI, add on therapy with memantine may be considered. If necessary, email dementia assessment service for advise).

3

Review

- **Review at 4 weeks, at 3 months, if well tolerated at yearly intervals** thereafter.
Check pulse (Follow flow chart page 5) patient tolerance, presence of side effects,(Table 2,3) **and carer's view on any improvement at each assessment. Use of objective clinical tools to assess cognition is not indicated.**
(Pulse check not required for memantine).
- **If side effects are severe and intolerable discontinue medication. Some side effects are dose dependent.** If side effects are not severe consider dose reduction first. **If necessary, email dementia assessment service for advise.**
- **If well tolerated, follow BNF dose increments** for individual drug **at each review.**
- If treatment is interrupted for longer than several days due to side effects or compliance, treatment should be re-initiated on the lower dose.
- Once established, continue medication long term as benefits can be minimal or hard to assess objectively in some cases. **Do not stop AChEI because of disease severity alone.**

Drug Information with oral use: Donepezil, Rivastigmine and Galantamine

See BNF for drug interactions and other side effects for individual drug

Table 1: Cautions in use: Donepezil, Rivastigmine and Galantamine

Concurrent condition	Effect
Cardiac disease, sick sinus syndrome, supraventricular conduction abnormalities. Do not prescribe for un-paced 2 nd or 3 rd degree heart block.	↑ Risk of bradycardia
Gastro or duodenal ulcers or susceptibility to ulcers	↑ Risk of ulcers
Asthma / COPD	↑ Risk of bronchospasm
History of convulsions	↑ Risk of generalised convulsions

Table 2: Potential few relevant side effects common to all AChEI

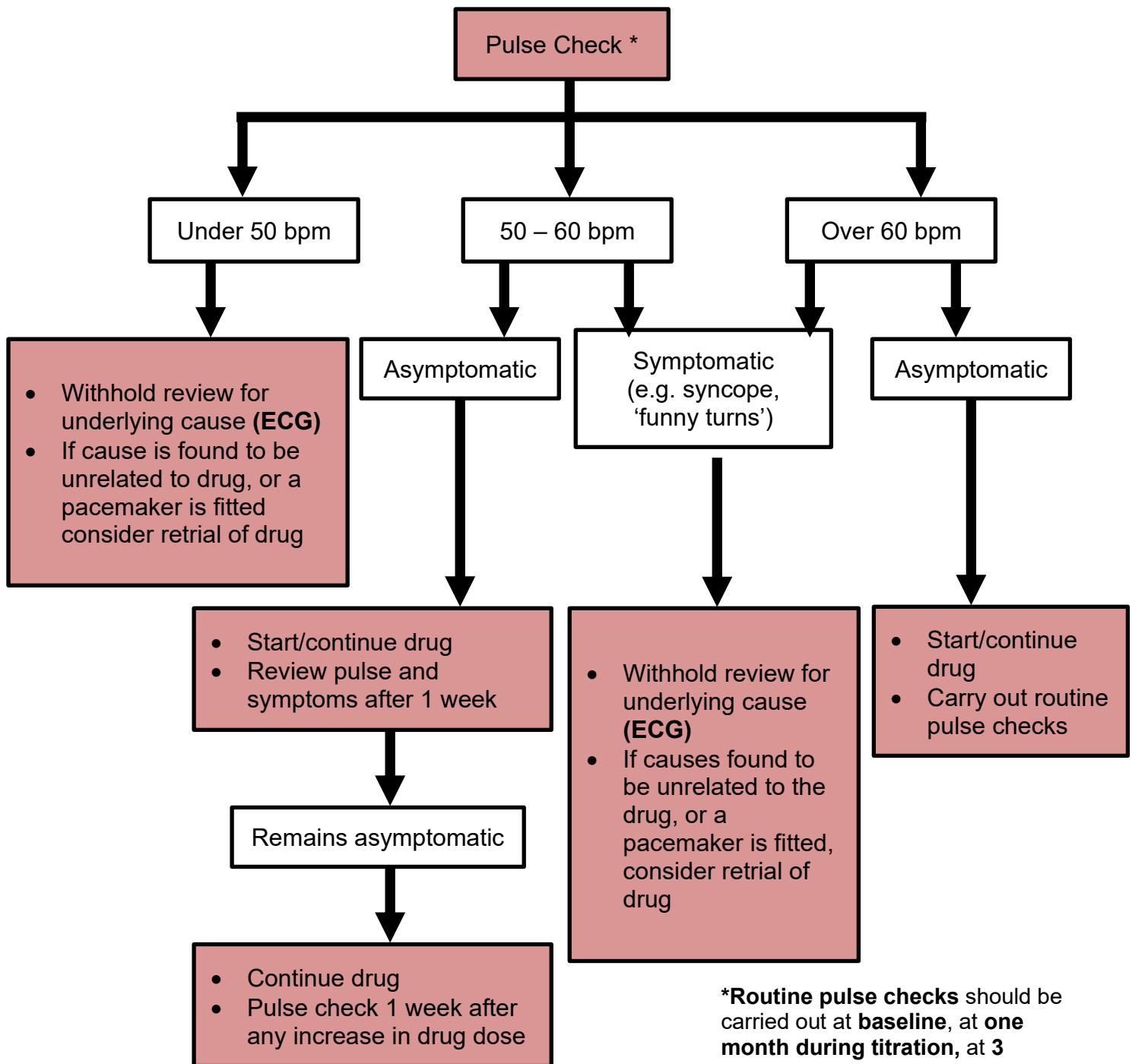
Frequency	Side effects
Common or very common	Diarrhoea, nausea, vomiting, headache, dizziness
Relatively more common	Arrhythmias with galantamine and rivastigmine than donepezil. Urinary incontinence more common with donepezil and rivastigmine.
Uncommon	AV block, bradycardia, gastric and duodenal ulcers and gastrointestinal haemorrhage
Rare	seizures
*Reports of GI side effects, hallucinations, abnormal dreams, nightmares, agitation and aggressive behaviour have resolved on dose-reduction or discontinuation of treatment of donepezil. **In investigating patients for syncope or seizure the possibility of heart block or long pauses should be considered. ***In cases of unexplained liver dysfunction, withdrawal of donepezil should be considered. ****Rare serious skin reactions reported with Galantamine	

Drug Information: Memantine

Table 3: Cautions and potential side effects of memantine

Frequency	Side effects
Common or very common	Balance impair., constipation, dizziness, drowsiness, dyspnoea, headache, hypersensitivity, hypertension
Uncommon	Confusion, embolism and thrombosis, fatigue, fungal infections, hallucination, heart failure, vomiting
Rare or very rare	Seizures
Frequency not known	Hepatitis, pancreatitis, psychotic disorders
*** Cautions : Epilepsy, history of convulsions, risk of convulsions.	

Adopted from Roland et al 2007, guidelines for managing cardiovascular risk prior to and during treatment with **acetylcholinesterase inhibitors** in dementia. bpm, heartbeats per minute; the 'drug' means the chosen AChE inhibitor



***Routine pulse checks** should be carried out at **baseline**, at **one month during titration**, at **3 months** and **yearly intervals** thereafter.

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